STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM

(SECMS)

HIGH-LEVEL PROJECT REPORT

Salesforce-Based Academic Enrollment Management Solution

Technology: Salesforce Lightning

Methodology: Agile / Sprint-Based Development

Prepared from the supplied SECMS project guide

PROJECT TEAM

Student Enrollment & Course Management System (SECMS)

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Role: Salesforce Lead Developer & Architect

Specialization: Custom Object Modeling, Salesforce Lightning Components, and Flow Automation

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Role: Salesforce Admin & Security Specialist

Specialization: Profiles, Permission Sets, Security Configurations, and Roll-up Summaries

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Role: Business Analyst & UI/UX Designer

Specialization: Customer Journey Mapping, User Stories, and Lightning App Page Layouts

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Role: Quality Assurance & Analytics Engineer

Specialization: Functional Testing, Reports, Dashboard Generation, and System Performance Validation

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Role: Salesforce Developer

Specialization: Apex, SOQL, and Lightning Web Components

Team responsibilities align with the SECMS Salesforce architecture, automation, security, user experience, development, testing, and reporting requirements described in the project guide.

EXECUTIVE SUMMARY

The Student Enrollment & Course Management System (SECMS) is a comprehensive Salesforce application designed to streamline and automate academic enrollment operations. It centralizes student records, course offerings, enrollments, instructor assignments, approvals, payments and reporting in one Salesforce environment. The guide defines the system as an end-to-end workflow from admission and course registration through approval, instructor assignment, revenue tracking and completion. ■filecite■turn0file0■L5-L24■

The project replaces manual and fragmented processes with structured workflows, validation rules, Salesforce Flows, Approval Processes, Reports, Dashboards and role-based security. Its stated purpose is accurate student data management, structured enrollment workflows, automated approvals and validations, and transparent monitoring. ■filecite■turn0file0■L25-L32■

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1. INTRODUCTION

1.1 Project Overview

SECMS centralizes student, course and enrollment data within a single Salesforce application. Enrollment Officers manage admissions, course registrations and communication, while instructors and administrators gain visibility into course allocations and academic progress. The guide emphasizes automated instructor assignment, task creation, validation and multi-level approval processes. ■filecite■turn0file0■L11-L24■

1.2 Purpose

- Maintain accurate student records.
- Create and manage course offerings.
- Process enrollment requests with validation and approval.
- Automate instructor assignment and follow-up.
- Track fee/payment status and revenue.
- Provide reports and dashboards.
- Protect academic information with role-based access.

1.3 Project Objectives

- Reduce duplicate data entry and fragmented records.
- Reduce manual approval and follow-up effort.
- Improve transparency of enrollment operations.
- Provide timely management reporting.
- Create a scalable cloud foundation for future modules.

2. IDEATION & PROBLEM ANALYSIS

2.1 Problem Statement

Persona	Goal	Problem	Impact
Enrollment Officer	Manage student enrollments	Scattered data; manual and unintegrated systems	Frustration and errors
Administrator	Track enrollments and revenue	Delayed, non-centralized reporting	Uncertain decision-making

2.2 Empathy Map

- Think/Feel: concern about double bookings and revenue loss.
- Hear: student and management complaints.
- See: multiple spreadsheets and emails.
- Say/Do: recheck information and follow up manually.
- Pain: duplicate entries and approval bottlenecks.
- Gain: automation, dashboards and faster processing.

These personas and pain points are directly reflected in the guide's empathy-map section. ■filecite■turn0file0■L68-L75■

2.3 Brainstorming & Prioritization

- Spreadsheet automation
- Standalone enrollment portal
- ERP integration
- Salesforce-based enrollment system

The guide groups solutions as manual, semi-automated and fully automated cloud solutions, then selects Salesforce-based SECMS because it was assessed as high feasibility and high impact. ■filecite■turn0file0■L77-L89■

2.4 Customer Journey

- Awareness & Registration → Course Selection → Enrollment Request → Fee Payment → Approval → Instructor Assignment → Reporting & Completion

3. REQUIREMENT ANALYSIS

3.1 Functional Requirements

ID	Epic	Requirement
FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve and reject enrollments
FR-4	Instructor Assignment	Automatically assign instructors
FR-5	Reporting	Provide dashboards and reports

The five functional requirements above are explicitly listed in the supplied guide. ■filecite■turn0file0■L101-L109■

3.2 Non-Functional Requirements

ID	Quality Requirement	Implementation Focus
NFR-1	Usability	Simple interface, clear tabs/forms, role-based dashboards
NFR-2	Security	Authentication, authorization and controlled access
NFR-3	Reliability	Avoid duplicates and maintain correct status/reporting
NFR-4	Performance	Fast search, dashboard refresh and bulk operations
NFR-5	Scalability	Support additional students, courses, departments and users
NFR-6	Availability	Continuous cloud access and minimized downtime

These six non-functional requirements and their examples are specified in the guide. ■filecite■turn0file0■L110-L160■

3.3 Data Flow

Users enter student and enrollment information; the system validates and stores records; approval processes route requests; instructor assignment automation processes approved data; reports and dashboards consume processed records. ■filecite■turn0file0■L162-L168■

4. SYSTEM DESIGN & ARCHITECTURE

4.1 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules + Approval Process
Reporting	Reports + Dashboards

This stack follows the guide's technology table. ■filecite■turn0file0■L169-L176■

4.2 Core Data Model

Object	Purpose	Main Fields / Links
Student	Student identity and contact	Name, Email, Phone, DOB, Status, Address
Course	Course offering	Name, Category, Duration, Fee, Description
Instructor	Instructor details	Name, Code, Expertise, Phone, Email
Enrollment	Central transaction	Student, Course, Instructor, Date, Status, Fees Paid, Amount, Comments

The guide specifies these objects and Enrollment lookup relationships to Student, Course and Instructor. ■filecite■turn0file0■L1467-L1490■ ■filecite■turn0file0■L1528-L1540■

4.3 Relationship Model

- Student 1:N Enrollment
- Course 1:N Enrollment
- Instructor 1:N Enrollment
- Enrollment is the central transaction connecting academic participants.

4.4 Logical Architecture

User → Lightning App → Objects → Validation / Flow → Approval → Instructor Assignment → Reports & Dashboards

5. PROJECT PLANNING & SCHEDULING

5.1 Agile Methodology

The guide uses Agile methodology with Epics → Stories → Story Points, sprint-based execution and velocity-based estimation. ■filecite■turn0file0■L195-L201■

Sprint	Scope	Points	Duration
1	Developer setup + initial data modeling	20	6 days
2	Data modeling, fields, relationships, navigation	20	6 days
3	Automation, validation and approval	20	6 days
4	Security	20	6 days
5	Reports	Backlog	Planned
6	Dashboards	Backlog	Planned

The first four sprints and their 20-point / six-day structure are documented in the guide; later backlog items cover reporting and dashboards. ■filecite■turn0file0■L1373-L1392■

5.2 Representative User Stories

ID	User Story
USN-1	Create a Salesforce developer account for project configuration.
USN-2	Create Student, Course, Enrollment and Instructor custom objects.
USN-4	Create fields for complete student, course and enrollment data.
USN-8	Validate mandatory data and enrollment constraints.
USN-9	Automate enrollment processing and notifications.
USN-10	Approve or reject enrollment requests before confirmation.
USN-11	Configure role-based security.
USN-12	Create reports for enrollments, courses and instructors.
USN-13	Create visual dashboards for statistics and trends.

6. SALESFORCE DEVELOPMENT & IMPLEMENTATION

6.1 Development Milestones

Milestone	Activity
1	Salesforce Developer Account Creation
2	Custom Objects: Student, Course, Instructor, Enrollment
3	Tabs Creation
4	Lightning App Creation
5	Fields Creation
6	Relationships
7	Page Layout Customization
8	Validation Rules
9	Flows / Automation
10	Approval Process
11	Reports
12	Dashboards
13	Security Setup
14	Conclusion

The milestone sequence follows the supplied guide. ■filecite■turn0file0■L1393-L1409■

6.2 Student Object

Create a Student custom object to store personal, contact and enrollment-status information. The guide specifies Student Name, Email, Phone, Date of Birth, Student Status and Address as core fields, with reports/search support enabled. ■filecite■turn0file0■L1467-L1488■

6.3 Course Object

Course contains Course Name, Category, Duration, Course Fee and Description. These fields support course catalog management and downstream enrollment and revenue reporting. ■filecite■turn0file0■L1490-L1504■

6.4 Instructor Object

Instructor contains name, instructor code, expertise, phone and email. The code can support the automated category-to-instructor assignment flow. ■filecite■turn0file0■L1514-L1525■

6.5 Enrollment Object

Enrollment contains Student, Course and Instructor lookup fields, Enrollment Date, Enrollment Status, Fees Paid, Total Amount and Comments. ■filecite■turn0file0■L1528-L1556■

7. AUTOMATION & APPROVAL

7.1 Relationships

Enrollment-to-Student, Enrollment-to-Course and Enrollment-to-Instructor lookup relationships provide the central data model. The guide gives step-by-step Salesforce configuration for each lookup. ■filecite■turn0file0■L1542-L1582■

7.2 Validation Rules

Rule	Purpose	Example
Student Email Domain	Ensure institutional email	NOT(CONTAINS(Email__c, "@student.edu"))
Fees Before Approval	Block approval when unpaid	Approved AND Fees_Paid__c = FALSE

These examples are specified in the guide. ■filecite■turn0file0■L1584-L1612■

7.3 Approval Process

Enrollment requests enter approval when Enrollment_Status__c = Requested. A Training Manager is the approver. Final approval sets the status to Approved; rejection sets it to Rejected and may send an email notification to the student. ■filecite■turn0file0■L1614-L1636■

7.4 Record-Triggered Flows

Flow	Trigger	Action
SetEnrollmentDate	Enrollment created	Set Enrollment Date to current date
Enrollment Approved Action	Enrollment updated to Approved	Email student; optionally notify admin; set Student status Active
AutoAssign_Instructor	Enrollment created/updated	Get Course; decide by category; assign instructor
Create_Followup_Task	Enrollment status Requested	Create admin follow-up task due Today + 2

These four automation scenarios are documented in the guide. ■filecite■turn0file0■L1737-L1808■

8. SECURITY DESIGN

8.1 Roles

Role	Access Model
Admin	Full access and configuration
Enrollment Officer	Create/Update Enrollment; read Student/Course as required
Instructor	Read-only access to assigned Enrollment records

These profiles and example permissions are specified by the guide. ■filecite■turn0file0■L1851-L1869■

8.2 Permission Sets

The guide proposes Custom Object Access and Dashboard Editor permission sets for additional access to project objects and reporting capabilities. ■filecite■turn0file0■L1870-L1876■

8.3 Organization-Wide Defaults

Object	OWD in Guide
Student	Public Read/Write
Course	Public Read/Write
Enrollment	Private

For instructor-specific access, the guide suggests an Instructor User lookup and a flow that can set OwnerId to the instructor user. ■filecite■turn0file0■L1877-L1891■

8.4 Security Principles

- Least privilege
- Role-based authorization
- Controlled access to student data
- Private enrollment transactions
- Auditable workflow transitions

9. REPORTS & DASHBOARDS

9.1 Reports

Report	Purpose
Students by Status	Group Student records by Status and display contact details.
Enrollments by Course	Group enrollment records by Course and show record count.
Pending Enrollment Approvals	Filter Enrollment Status = Requested.
Revenue Report – Fees Paid	Filter Fees Paid = TRUE and summarize fees by course.

These reports and their filters/grouping are directly described in the guide. ■filecite■turn0file0■L1809-L1833■

9.2 Dashboard

- Donut Chart → Students by Status
- Bar Chart → Enrollments by Course
- Gauge → Revenue / Fees Paid
- Table → Pending Enrollment Approvals

The dashboard provides a consolidated operational view of student status, enrollment volume, revenue and pending approvals. ■filecite■turn0file0■L1835-L1850■

9.3 Recommended KPI Definitions

KPI	Definition
Total Students	Count of Student records
Active Students	Students with Active status
Total Enrollments	Count of Enrollment records
Pending Approvals	Enrollments with Requested status
Paid Revenue	Sum of fees where Fees Paid = TRUE
Enrollment by Course	Enrollment count grouped by Course

10. TESTING & QUALITY ASSURANCE

10.1 Testing Scope

The guide requires functional and performance testing covering record creation, validation rules, record types, flow execution, approval and seat-availability checks, Apex-trigger functionality where implemented, reports and dashboards. Screenshots should be captured as testing evidence. ■filecite■turn0file0■L1892-L1902■

10.2 Test Case Matrix

ID	Test	Expected Result
TC-01	Create valid Student	Record saved
TC-02	Invalid institutional email	Validation error
TC-03	Incomplete Enrollment	Validation blocks save
TC-04	Requested Enrollment	Approval process entry criteria met
TC-05	Approve unpaid Enrollment	Fee validation prevents approval
TC-06	Approve paid Enrollment	Status becomes Approved
TC-07	Approved Enrollment	Student becomes Active through automation
TC-08	Technical Course enrollment	Technical instructor assigned
TC-09	Requested Enrollment	Follow-up Task created
TC-10	Open dashboard	Charts and table reflect report data

10.3 Acceptance Criteria

- Core objects can be created by authorized users.
- Relationships correctly link records.
- Invalid records are blocked.
- Approval status changes are controlled.
- Configured flows execute successfully.
- Reports and dashboards match source records.
- Restricted enrollment data is not exposed to unauthorized roles.

11. ADVANTAGES & LIMITATIONS

Advantages	Limitations / Considerations
Centralized data	Salesforce dependency
Automated workflows	Initial configuration effort
Real-time reporting	Requires governance and correct configuration
Role-based access	Users require suitable training
Scalable cloud foundation	Future integrations require additional design

The supplied guide explicitly identifies centralized data, automated workflows and real-time reporting as advantages, and Salesforce dependency and initial configuration effort as disadvantages. ■filecite■turn0file0■L1904-L1911■

11.1 Operational Benefits

- Faster enrollment processing
- Reduced manual follow-up
- Improved record consistency
- Better management visibility
- Structured instructor assignment
- Centralized reporting

12. CONCLUSION

SECMS provides an end-to-end Salesforce workflow for student registration, course creation, enrollment, instructor assignment, approval, revenue tracking and reporting. By using Salesforce validation rules, flows, approval processes, dashboards and security, the project converts fragmented academic processes into a structured digital solution. The guide concludes that the system improves data accuracy, operational efficiency and decision-making while remaining scalable, reliable and user-friendly. ■filecite■turn0file0■L1912-L1920■

13. FUTURE SCOPE

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

These four future enhancements are explicitly identified in the supplied guide. ■filecite■turn0file0■L1921-L1926■

13.1 Enhancement Roadmap

Phase	Enhancement	Target Capability
1	Payment Gateway	Online payment and fee synchronization
2	Student Portal	Self-service application, course selection and tracking
3	Advanced Analytics	Trend analysis and data-driven insights
4	Mobile Support	Mobile academic operations and notifications

14. APPENDIX – PROJECT DEMO & SUBMISSION CHECKLIST

#	Deliverable / Evidence
1	Salesforce Developer Org created
2	Student, Course, Instructor and Enrollment objects
3	Required fields configured
4	Lookup relationships configured
5	Tabs and Lightning application
6	New Enrollment and Re-Enrollment record types
7	Page layouts and related lists
8	Validation rules
9	Enrollment approval process
10	Enrollment date flow
11	Approval notification flow
12	Instructor assignment flow
13	Follow-up task flow
14	Profiles, permission sets and sharing
15	Operational reports
16	Student Management Dashboard
17	Functional/performance testing
18	Screenshots of testing and final UI
19	Project conclusion and future scope

14.1 Project Artifacts

- Source / Configuration: Salesforce configuration
- Dataset: Salesforce Objects
- Demo: Salesforce Developer Org
- Testing Evidence: Salesforce screenshots

The guide identifies Salesforce Configuration, Salesforce Objects and Salesforce Developer Org as the appendix artifacts. ■filecite■turn0file0■L1927-L1930■

— END OF PROJECT REPORT —